Company Property

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CI7625tg

3G AFE – Analog Baseband

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Table of Contents

1. Features

- 0.18µm standard CMOS
- Dual Voltage supply:
  - Analog 2.8V±10% operation
  - Digital 1.8V±10% operation
  - I/O 2.8V±10% operation
- I/Q Rx path
  - Two fully parallel paths for diversity
  - Supports CDMA/WCDMA/EGPRS standards
  - Programmable gain to optimize dynamic range
  - 12bit Sigma-Delta ADCs with typical 72dB dynamic range for CDMA/WCDMA and 80dB for EGPRS
- I/Q Tx path
  - Supports CDMA/WCDMA/EGPRS standards
  - 10bit DAC with Analog low-pass filtering
  - Supports both current and voltage output modes
- Auxiliary general purpose ADC
  - 10bit 400kS/s for
  - 6 multiplexed inputs
- Auxiliary Radio Control DACs
  - 12bit 40µs for AFC
  - 10bit 10µs for AGC
  - 10bit 10µs for general purpose
  - 10bit 0.5µs for PA control
- Internal reference generator including
  - Bandgap voltage
  - Reference current
- Typical Current consumption
  - Each I/Q Rx analog path (with PLL):
    - EGRPS: 7.5 mA
    - CDMA: 7.75 mA (2 PLLs active)
    - WCDMA: 12 mA
  - I/Q Tx analog path:
    - EGRPS voltage output: 10mA
    - CDMA voltage output: 10mA
    - WCDMA voltage output: 20mA
    - Current output: 8mA
  - Auxiliary converters (total): 2.5mA
  - Digital section < 12mA (WCDMA)
- Area (with I/O pads) < 12.5 mm^2
- Area (without I/O pads) < 9.5 mm^2
- Application: CDMA/WCDMA/EGPRS portable wireless units

2. General Description

This device is a fully integrated baseband analog processor that is capable of performing all the functions necessary for the DSP/Radio interface in EGPRS, CDMA and WCDMA applications.

The Radio Control section includes several auxiliary DACs for DSP/Radio interface in GSM/CDMA/WCDMA applications.

The device has an on-chip 10-bit Analog-to-digital converter, which is intended to digitize DC signals corresponding to analog parameters such as battery voltage, temperature, etc.

This device can operate from a power supply voltage ranging from 2.5V to 3V and is controlled by a digital baseband modem to maintain minimum power consumption.

All necessary voltage references and biasing currents are generated internally with only the need of some extra pins for decoupling.

A power down mode is available with less than 15µA current consumption.

3. Functional Diagram

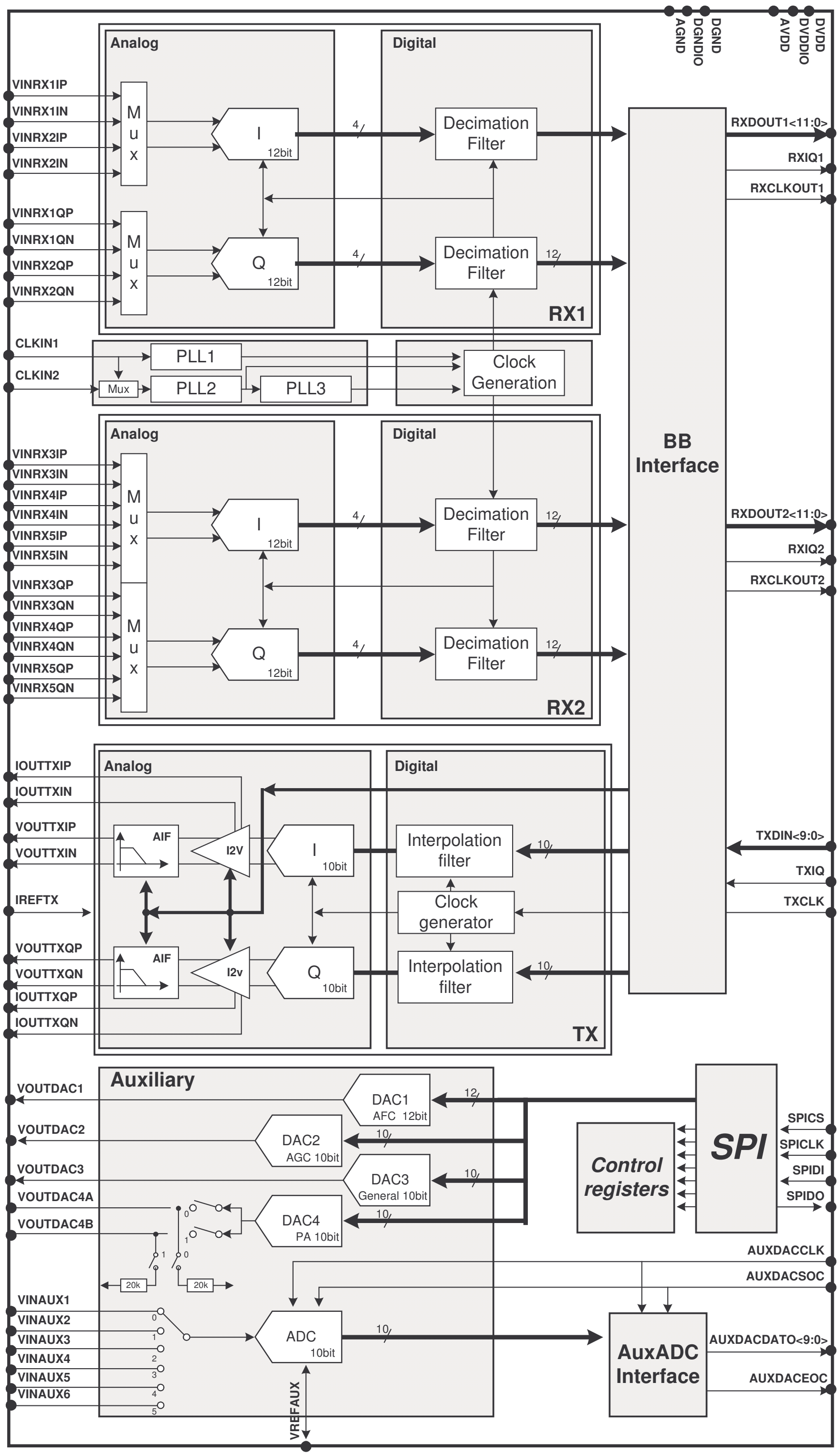

Figure 1 – Functional Diagram

4. Specifications

Typical values are at TA = 25ªC.
Minimum and maximum values apply over the full ambient temperature range.

Parameter	Conditions	MIN	TYP	MAX	Unit
Technology parameters					
Technology			0.18µm logic, 1P5M		
Absolute Maximum Ratings					
Analog supply voltage		-0.5		3.3	V
Digital supply voltage		-0.5		2	V
Analog Input Voltage		-0.5		3.5	V
Storage Temperature		-65		125	ºC
Max Junction Temperature				150	ºC
Temperature ranges					
Ambient Temperature, TA		- 40		+ 85	C
Junction Temperature, TJ		- 40	25	+ 125	C
System Supplies					
Digital Power Supply		1.6	1.8	2.0	V
Digital IO Power Supply		1.6	1.8	2.0	V
Analog Core Power Supply		2.5	2.7	3.0	V
Analog Interface Power Supply		2.5	2.7	3.0	V
Internal Voltage Reference	On AVDDREG pin	1.61	1.8	2.07	V
Logic Input & Output Signals					
VIH High Level Input Voltage		0.7*DVDD		DVDD+0.3	V
VIL Low Level Input Voltage		- 0.3		0.3*DVDD	V
VOH High Level Output Voltage		0.8*DVDD		DVDD	V
VOL Low Level Output Voltage		0		0.2*DVDD	V
Input current	High and low state	- 10		+ 10	uA
Mode Recovery Timing					
Power-Down Wake-Up Time	To idle mode (W/o taking PLL startup into account)			50	us
Switch Time from Idle	To TX Active	1		3	us
Switch Time from Idle	To RX Active			10	us
Rx I/Q Sigma-Delta ADC's					
Differential Input Voltage		0.4	0.6	1.2	Vrms
Common Mode Input Voltage		1	AVDD/2	2	V
Gain	Steps are: 0.5x 0.75x 1x 1.5x	-6		+3.5	dB
Input Resistance		20			kOhm
Input Capacitance				10	pF
Gain imbalance	Between I&Q ADC's		+/- 0.1		dB
Logic coding			2's complement		
Current consumption - Analog Active	Both RX paths (4 ADCs) EGPRS (13MHz) CDMA (39.3216MHz) WCDMA (61.44MHz)		 7.5 7.5 12		 mA mA mA

Current consumption - Digital Active (Decimation filter + Logic)	EGPRS ZIF Fclkout =52MHz Fclkout =26MHz Fclkout =13MHz EGPRS NZIF Fclkout =52MHz Fclkout =26MHz Fclkout =13MHz CDMA Fclkout =39.3216MHz Fclkout =19.6608MHz Fclkout =9.8304MHz WCDMA Fclkout =61.44MHz Fclkout =30.72MHz		 0.75 0.60 0.53 1 0.85 0.80 1.3 1.2 1.15 2 1.8		 mA mA mA mA mA mA mA mA mA mA mA
Current consumption - Total power-down			5		uA

EGPRS Mode

Sampling Rate	Zero-IF Low-IF		13 26		MHz MHz
Signal Bandwidth	Zero-IF Low-IF		100 200		KHz KHz
Dynamic Range	DR Zero-IF Low-IF		 80 77		 DB DB
Harmonic Distortion	THD		-80		DB
Digital filter (Decimation)					
Sampling Rate	Chipx4 ZIF NZIF		 1083.33 2166.66		kHz

CDMA Mode

Sampling Rate			39.3216		MHz
Signal Bandwidth			614.4		kHz
Dynamic Range	DR		72		dB
Harmonic Distortion	THD	-80			dB
Digital filter (Decimation)					
Sampling Rate	chipx4		4.9152		MHz
Resolution			12		bit

WCDMA Mode

Sampling Rate			61.44		MHz
Signal Bandwidth			1.92		MHz
Dynamic Range	DR		72		dB
Harmonic Distortion	THD	-70			dB
Digital filter (Decimation)					
Sampling Rate	chipx4		15.36		MHz
Resolution			12		bit

Tx I/Q DAC + Analog Filter

Resolution			10		bit
Differential non-linearity	DNL	-1		1	LSB
Integral non-linearity	INL	-1		1	LSB
Signal to Noise and Distortion	SINAD Integrated in the signal bandwidth		55		dB
Full scale error		-5		+5	%
Differential Output Voltage	Programmable	0.9		2	Vpp
Common Mode Output Voltage	Programmable	1		1.5	V
Offset calibration cycle duration				2000	cycles
Voltage Offset each I&Q channel	Before calibration After calibration	-25 -2		25 2	mV
Voltage Offset between I&Q	After calibration	-6		6	mV
Output Load	Referred to ground	25			kOhm
I&Q Gain mismatch		-0.2		+0.2	dB
Differential Capacitive Load				15	pF
Single-ended Capacitive Load				10	pF

Differential Resistive Load		10			kOhm
Logic coding			2's complement		
Current consumption - Analog Active	Voltage output mode EGPRS CDMA WCDMA		 10 10 20		 mA mA mA
Current consumption - Analog Active	Current output mode IREF=300uA EGPRS CDMA WCDMA		 8 8 8		 mA mA mA
Current consumption - Digital Active	EGPRS CDMA Fin=19.6608MHz Fin=9.8304MHz Fin=4.9152MHz WCDMA Fin=30.72MHz Fin=15.36MHz		0.45 1.5 0.85 0.8 2.4 2.1		mA mA mA mA mA mA
Current consumption - Total power-down	EGPRS CDMA WCDMA		5		uA
Current outputs					
INL In Current Mode	Iref from 20 to 300uA	-4		4	LSB
Current reference	External	20		300	uA
Differential Output Current (pk-pk)	Scales linearly with Iref Iref = 250uA ; Gain=0 Iref = 250uA ; Gain=1			 4 5	 mA mA
Common Mode Output Current	Scales linearly with Iref Iref = 250uA ; Gain=0 Iref = 250uA ; Gain=1			 1 1.25	 mA mA
Current Offset each I&Q channel		-15		15	uA
Current Offset between I&Q		-22		22	uA uA
Output Compliance Voltage		1.05		1.3	V
Current reference pin compliance Voltage				0.5	V
Voltage outputs - EGPRS Mode					
Sampling Rate			13		MHz
Attenuation at Sampling rate	@Fs = 13MHz	50			dB
Group delay deviation	From constant			100	ns
I&Q phase mismatch				1	Degr.
Filter Ripple Frequency Range		0		100	kHz
Filter Ripple		-0.2		0.2	dB
Voltage outputs - CDMA Mode					
Sampling Rate			39.3216		MHz
Attenuation at Sampling rate	@Fs = 39.3216MHz	50			dB
Group delay deviation	From constant			15	ns
I&Q phase mismatch				1	Degr.
Filter Ripple Frequency Range		0		614.4	kHz
Filter Ripple		-0.2		0.2	dB
Voltage outputs - WCDMA Mode					
Sampling Rate			61.44		MHz
Attenuation at Sampling rate	@Fs = 61.44MHz	50			dB
Group delay deviation	From constant			15	ns
I&Q phase mismatch				1	Degr.
Filter Ripple Frequency Range		0		1.92	MHz
Filter Ripple		-0.2		0.2	dB

Rx AGC, TX APC and GP Auxiliary DAC

Resolution			10		bit
Differential non-linearity	DNL	-1		1	LSB
Integral non-linearity	INL	1		1	LSB
SINAD	10kHz, full-scale input		50		dB
Settling Time	To 1% of final value			0.5	usec
Output Voltage	Referenced to bandgap	0		2.25	V

DC current to charge the output load (sourcing capability)				2.5	mA
DC current to discharge the output load (sinking capability)				0.4	mA
Load capacitance				20	pF
Current consumption	Normal operation mode Power down mode		560	700 0.5	uA uA
Logic coding			Unsigned binary		

AFC Auxiliary DAC

Resolution			12		bit
Differential non-linearity **(1)**	DNL	-1		1	LSB
Integral non-linearity **(1)**	INL	-4		4	LSB
SINAD **(2)**	10kHz, full-scale input		66		dB
Settling Time	To 1% of final value			40	usec
Output Voltage	Referenced to bandgap	0		2.25	V
DC current to charge the output load (sourcing capability)				2.5	mA
DC current to discharge the output load (sinking capability)				0.4	mA
Load capacitance				20	pF
Current consumption	Normal operation mode Power down mode		610	750 0.5	uA uA
Logic coding			Unsigned binary		

Auxiliary ADC

Resolution			10		bit
Maximum Sample Rate				400	kHz
Latency			10		cycles
Differential non-linearity	DNL	-1		1	LSB
Integral non-linearity	INL	-1		1	LSB
DC Offset Error		-20		20	mV
Input Swing	Internal reference mode External reference mode	0 0		AVDD VREFAUX	V
External reference range	External reference mode	2		AVDD	V
Input Resistance			100		kOhm
Input Capacitance			20		pF
Current consumption	Normal operation mode Power down mode			400 0.5	uA uA
Logic coding			Unsigned binary		

PLLs

Input Reference Frequency		13		26	MHz
Peak-to-peak, cycle-to-cycle jitter	long-term, rms		60		ps
Lock Time	Fcomparison=400kHz Fcomparison=13MHz			500 16	us
Startup time	After reset			500	us
Current consumption	Normal operation mode Fvco=250MHz Fvco=600MHz Power down mode		 1.5 3 	 1	 mA mA uA

References

Power consumption	Bandgap + LDO Power down mode		300	400 0.5	uA uA
Power up time	Bandgap + LDO			40	us

Note:

(1) Due to saturation of the output buffer, INL and DNL are not applicable for output voltages below 250mV and above AVDD-250mV. Output is monotonic above 0.15V.

(2) SINAD spec does not account for the INL degradation

5. Pin Description

Pin Name	I/O	Type	Function
SUPPLIES			
DVDD	I/O	Supply	**1.8V** digital supply
DVDDIO	I/O	Supply	**1.8V** I/O ring digital supply
DGND	I/O	Supply	I/O ring digital ground
DGNDIO	I/O	Supply	**1.8V** I/O ring digital supply
AVDD	I/O	Supply	**2.7V** analog supply
AGND	I/O	Supply	Analog ground
AGNDREF	I/O	Supply	Clean reference signal ground
REFERENCES			
AVDDREG	I/O	Analog	**1.8V** internal supply voltage for off-chip decoupling. Connect **1µF** in parallel with **100nF** ceramic capacitors to agnd.
VREFAFE	I/O	Analog	AFE reference voltage for off-chip decoupling. Connect **1µF** in parallel with **100nF** ceramic capacitors to agndref
VREFAUX	I	Analog	Reference voltage for AuxADC (when in External Reference Mode)
VTESTP	I/O	Analog	Analog test pin (either positive side on differential signals or single-ended – depends on the chosen internal test mode)
VTESTN	I/O	Analog	Analog test pin (either negative side on differential signals or single-ended – depends on the chosen internal test mode)
RX			
VINRX1IP	I	Analog	Rx1 I-channel Positive voltage input signal
VINRX1IN	I	Analog	Rx1 I-channel Negative voltage input signal
VINRX1QP	I	Analog	Rx1 Q-channel Positive voltage input signal
VINRX1QN	I	Analog	Rx1 Q-channel Negative voltage input signal
VINRX2IP	I	Analog	Rx2 I-channel Positive voltage input signal
VINRX2IN	I	Analog	Rx2 I-channel Negative voltage input signal
VINRX2QP	I	Analog	Rx2 Q-channel Positive voltage input signal
VINRX2QN	I	Analog	Rx2 Q-channel Negative voltage input signal
VINRX3IP	I	Analog	Rx3 I-channel Positive voltage input signal
VINRX3IN	I	Analog	Rx3 I-channel Negative voltage input signal
VINRX3QP	I	Analog	Rx3 Q-channel Positive voltage input signal
VINRX3QN	I	Analog	Rx3 Q-channel Negative voltage input signal
VINRX4IP	I	Analog	Rx4 I-channel Positive voltage input signal
VINRX4IN	I	Analog	Rx4 I-channel Negative voltage input signal
VINRX4QP	I	Analog	Rx4 Q-channel Positive voltage input signal
VINRX4QN	I	Analog	Rx4 Q-channel Negative voltage input signal
VINRX5IP	I	Analog	Rx5 I-channel Positive voltage input signal
VINRX5IN	I	Analog	Rx5 I-channel Negative voltage input signal
VINRX5QP	I	Analog	Rx5 Q-channel Positive voltage input signal
VINRX5QN	I	Analog	Rx5 Q-channel Negative voltage input signal
TX			
IOUTTXIP	O	Analog	Tx I-channel Positive current output signal
IOUTTXIN	O	Analog	Tx I-channel Negative current output signal

IOUTTXQP	O	Analog	Tx Q-channel Positive current output signal
IOUTTXQN	O	Analog	Tx Q-channel Negative current output signal

VOUTTXIP	O	Analog	Tx I-channel Positive voltage output
VOUTTXIN	O	Analog	Tx I-channel Negative voltage output
VOUTTXQP	O	Analog	Tx Q-channel Positive voltage output
VOUTTXQN	O	Analog	Tx Q-channel Negative voltage output

IREFTX	I	Analog	External current reference signal. External current must be connected from this pin to ground.
AuxADC			
VINAUX1	I	Analog	Analog input 1
VINAUX2	I	Analog	Analog input 2
VINAUX3	I	Analog	Analog input 3
VINAUX4	I	Analog	Analog input 4
VINAUX5	I	Analog	Analog input 5
VINAUX6	I	Analog	Analog input 6
AuxDACs			
VOUTDAC1	O	Analog	DAC 1 output voltage. *AFCDAC*
VOUTDAC2	O	Analog	DAC 2 output voltage. *RX AGCDAC*
VOUTDAC3	O	Analog	DAC 3 output voltage. *GPDAC*
VOUTDAC4A	O	Analog	DAC 4 output voltage A. *TX APCDAC*
VOUTDAC4B	O	Analog	DAC 4 output voltage B. *TX APCDAC*
Clocks			
CLKIN1	I	Digital	Input reference clock 1
CLKIN2	I	Digital	Input reference clock 2
Data Interface			
RSTZ	I	Digital	Master reset. Asynchronous and active low
INT	O	Digital	Interrupt signal.

RXCLKOUT1	O	Digital	Rx1 output clock to baseband processor.
RXDOUT1<11:0>	O	Digital	Rx1 data output. I and Q are multiplexed on the same pins
RXIQ1	O	Digital	Rx1 Indicator of output data content. I sample on ‘1’, Q sample on ‘0’.

RXCLKOUT2	O	Digital	Rx2 output clock to baseband processor.
RXDOUT2<11:0>	O	Digital	Rx2 data output. I and Q are multiplexed on the same pins
RXIQ2	O	Digital	Rx2 Indicator of output data content. I sample on ‘1’, Q sample on ‘0’

TXCLK	I	Digital	Tx interface clock from baseband processor.
TXDIN<9:0>	I	Digital	Tx data input. I and Q are multiplexed on the same pins
TXIQ	I	Digital	Tx Indicator of input data content. I sample on ‘1’, Q sample on ‘0’.

AUXADCSOC	I	Digital	AUXADC start-of-conversion
AUXADCEOC	O	Digital	AUXADC end-of-conversion signal
AUXADCCLK	I	Digital	AUXADC input clock
AUXADCDATO	O	Digital	AUXADC output signal.

ENLDO	I	Digital	LDO enable
SCANMODE	I	Digital	SCAN mode enable.

SPICS	I	Digital	SPI Chip Select Signal
SPICLK	I	Digital	SPI Clock Signal
SPIDI	I	Digital	SPI Data In Signal
SPIDO	O	Digital	SPI Data Out Signal

Note:
All signals are active high unless where the opposite is stated.

6. Functional description

This device consists of:

- Two EGPRS / CDMA / WCDMA Rx path sections with input multiplexers,
- One EGPRS / CDMA / WCDMA Tx path,
- A general-purpose ADC with six multiplexed inputs,
- Four auxiliary DACs: for radio control (AFC, AGC, PA) and general purpose,
- Three PLLs for the Rx paths.

All needed reference voltages and currents are generated internally with only the need of two external pins for decoupling: one for the bandgap generator and the other one for an LDO that regulates a 1.8V internal supply voltage for the I/Q ADC, I/Q DAC, PLL and auxiliary converters.

All blocks can be powered-up/down independently.
In normal operation mode, after supply voltage is first applied, a master reset must be generated (***rstz="0"***) before the blocks are turned on.

6.1. RX path

6.1.1. Description

The EGPRS (ZIF/NZIF) / CDMA / WCDMA receive path interface is designed to accept differential I/Q baseband analog signals. These are converted by a ΣΔ ADC. The output bitstream is then passed through a decimation filter that reduces the data rate before being sent to the digital baseband modem where it is filtered by a very selective FIR filter.

There are two identical RX paths for diversity reception and each has an input multiplexer that enables the connection of a maximum of three RF devices.

The mode of operation is selected through SPI registers.

6.1.2. I/Q ADC

These are multi-standard, oversampled **12bit ΣΔ ADC**. The mode of operation is defined by changing the sampling frequency and by setting a SPI register. The sample rate is then:

Mode of operation	Sample rate
EGPRS ZIF	13 MHz
EGPRS NZIF	26 MHz
CDMA	39.3216 MHz
WCDMA	61.44 MHz

In order to maximize the dynamic range for any input signal level, the gain can be programmed through SPI register. Possible gain values are **0.5**, **0.75**, **1** and **1.5**. This range of gains guarantees that there is no significant loss of dynamic range for input voltages ranging from **0.8 Vrmsdif** to **1.2 Vrmsdif**.

For CDMA / WCDMA modes the ΣΔ ADC has a minimum **65dB** dynamic range in the channel bandwidth at **0dB** gain. For EGPRS the minimum dynamic range is **75dB**.
The gain does not need to be re-programmed to calibrate from one handset to the next.

A level shifter function is also included in order to make the ADC input common mode voltage independent of the RF device output common mode voltage.

In order to enable simultaneously connection between the AFE a several RF devices, a 3:1 multiplexer is available at the ADC input.

The ADC reference voltage is generated internally from the 1.8V analog supply voltage without the need of any extra pins.

6.1.3. Decimator

The decimation filter reduces the data rate to **chipX4** (**4.9152MHz** for CDMA / **15.36MHz** for WCDMA and **1.083MHz (ZIF)**/**2.166MHz (NZIF)** for EGPRS) before being sent to the digital baseband modem. This filter provides **12**-**bit** output. Alternatively a **16-bit** output is available by using the data interface at double-rate.

6.1.4. Power modes

Each Rx path can be set independently into one of the following states:

- Power-on – all internal blocks are turned on
- Power-down – all internal blocks are turned off and power consumption is reduced to the minimum.

6.2. TX path

6.2.1. Description

The EGPRS / CDMA / WCDMA transmit path is designed to accept I/Q baseband digital signals at a programmable update rate of **chipx4**, **chipx8** or **chipx16** for CDMA, **chipx4** or **chipx8** for WCDMA and **chipx4** for EGPRS mode.

In order to relax the analog anti-aliasing filter requirements, the signal is first passed through an interpolator, before being applied to an I/Q DAC. The output can be in current, or voltage mode. On the last case an Anti-Imaging Filter (AIF) attenuates the out of band signal image and performs current to voltage (I2V) conversion.

The mode of operation is selected through SPI registers.

6.2.2. Interpolator filter

The input signals are first passed through an interpolator that increases the sample rate to **chipx32 (39.3216MHz)** for CDMA, **chipx16 (61.44MHz)** for WCDMA and **chipx48 (13MHz)** for EGPRS.

6.2.3. I/Q DAC

This is a **10 bit** current-steering I/Q DAC working at a clock speed higher than the chip rate as shown below:

Mode of operation	Sample rate - Fs
EGPRS	13 MHz
CDMA	39.3216 MHz
WCDMA	61.44 MHz

6.2.4. Current mode output

On this mode, an external current reference applied through IREFTX is used to define the output level. The DAC resolution is always guaranteed to be better than **8 bit**. The current reference ranges between **20µA** and **300µA**.

The output current full-scale level scales linearly with the current reference and can be programmed to be **4.0mAdiff pk-pk** or **5mAdiff pk-pk** (when **IREF=250uA**) through a SPI register.
Maximum compliance voltage is **1.2V** for **4.0mAdiff** output and **1.3V** for **5mAdiff** output.

When in current output mode, the AIF can be turned off, to save power consumption

6.2.5. Voltage mode output - Analog Anti-Imaging Filter (AIF)

When in voltage mode, an analog filter provides enough rejection of the signal image around the DAC sampling clock and performs current to voltage (I2V) conversion

To improve the image rejection with in-band ripple less than **0.2dB**, the cut-off frequency and power-consumption is optimized for each mode of operation:

Mode of operation	Maximum Inband ripple (dB)	Group Delay deviation from constant (ns)	Image rejection@Fs (dB)
EGPRS	**+- 0.2**	**100**	**>50**
CDMA	**+- 0.2**	**15**	**>50**
WCDMA	**+- 0.2**	**15**	**>50**

The filter output voltage is programmable from **0.9V** to **2V** with 15 gain steps.

The output common-mode voltage is also programmable from **1V** to **1.5V**.

6.2.6. Offset calibration

An internal offset calibration function is available to compensate for the differential offset of the complete Tx channel and then the common mode offset between I and Q channels. The compensation has to be re-processed each time the mode is changed.

In voltage output mode, the calibration machine measures the voltage output pins VOUTTXIP, VOUTTXIN, VOUTTXQP, VOUTTXQN.

In current output mode, the calibration machine measures the current output pins IOUTTXIP, IOUTTXIN, IOUTTXQP, IOUTTXQN. Since the calibration machine measures voltages (not currents), this calibration is only applicable if I2V is performed directly at the output pins (through a resistor connected to ground). Otherwise, calibration must be performed with an external loop. The offset on the current output mode is very low, being less than **±15µA**.

The calibration process lasts less than **2000** TX clock cycles. The remaining differential offset is **+-1LSB** and the I/Q offset error is less than **+-2mV**. The current offset, measured as a percentage of the full scale, is equivalent to the voltage offset.

Start of calibration is performed through the SPI, and after calibration has finished, an end-of-calibration is signaled through the SPI.

Since for each standard, the filters specifications change, each must be calibrated independently.

6.2.7. Power modes

The Tx path can be set into one of the following states:

- Power-on – all internal blocks are turned on
- Power-down – all internal blocks are turned off and power consumption is reduced to the minimum. When in power-down mode, calibration value is not reset. To do so a master reset (rstz='0') must be performed.
  All voltage outputs, current outputs and current reference input stay at high impedance when in power-down.

6.3. General purpose ADC (AuxADC)

6.3.1. Description

This is a general purpose ADC that can be used for housekeeping tasks.
There are **6** multiplexed inputs that can be selectable. These enable the monitoring of several signals on the front-end, such as battery voltage or temperature, without adding to the overall power consumption.

The ADC is a **10bit** SAR with **400kS/sec** maximum sample rate. The reference voltage is derived either directly from the supply (**2.8V±10%**) or from an external source through pin ***vrefaux*** (minimum reference value is **2V**).

Data latency is equal to **12 cycles.** Clock frequency for the conversion is at a frequency of no higher than **4.8MHz**

All data transfers are done through a dedicated Serial interface. The inputs and modes of operation are selectable through an SPI register**.**

6.3.2. Power modes

The AuxADC can be set into one of the following states:

- Power-on – all internal blocks are turned on
- Power-down – all internal blocks are turned off and power consumption is reduced to the minimum.

6.4. Radio control DACs (AuxDACs)

There are 4 Auxiliary DACs provided for the following functions:

- 12bit DAC, 40µs for Automatic Frequency Correction (AFC)
- 10bit DAC, 10µs for Automatic Gain Control (AGC)
- 10bit DAC, 0.5µs for other functions (including APC)
- 10bit DAC, 10µs for general purposes

6.4.1. Reference

The reference voltage is derived directly from the bandgap generator. The output full-scale is **2.25V**.

6.4.2. Automatic Frequency Correction DAC (DAC1)

This **12bit** DAC with **40µs** settling time is required to tune a VCO reference, where used and where not controlled directly by a RF transceiver.

The architecture is comprised of a digital sigma-delta truncator that truncates from 12 to 10 bits, and a Current-switched 10bit DAC with I2V output conversion and filtering (to reduce ripple). The sample rate at the DAC input is divided from one of the Rx PLL output clocks and the division factor is programmable through a SPI register. To guarantee the required SINAD and DNL spec, the clock must be guaranteed to be always higher than **6.5MHz**.

Operation mode is set through a SPI register.

6.4.3. Automatic Gain Control DAC (DAC2)

This **10bit** current-switched DAC with **0.5µs** settling time is required to control automatically the receive gain in the RF transceiver.
Operation mode is set through a SPI register.

6.4.4. General Purpose DAC (DAC3)

This is a general purpose **10bit** current-switched DAC with **0.5µs** settling time which can be used for LNA control or PA control.
Operation mode is set through a SPI register.

6.4.5. Automatic Power Control DAC (DAC4)

This **10bit** current-switched DAC with **0.5µs** settling time can be used to control the transmit power in the RF transceiver, including PA ramp requirements for GSM systems.

The output of this DAC can be passed to one of two outputs, depending on the setting of an internal analogue multiplexer. Both outputs are pulled down through a nominally **20kΩ** resistor after the switch. This allows the single DAC to be used for both EGPRS and WCDMA PA control. Only one output can be enabled at the same time, otherwise the output may become instable and the linearity seriously degraded.

All possible modes of operation are set through a SPI register.

The DAC operation mode can be either selected by the ramp generator or by direct setting of SPI registers. The following table details the mode selection.

Ramp enable (SPI) = 0 DAC4 enable (SPI) = 0	Ramp enable (SPI) = 1 DAC4 enable (SPI) = x	Ramp enable (SPI) = 0 DAC4 enable (SPI) = 1
Output enable = 00	Output enable = ramp timing AND Output enable_SPI	Output enable = Output enable control_SPI
Output pull-down = 11	Output enable = ramp timing OR Output pull-down control_SPI	Output power-down = Output pull-down control_SPI

The "SPI" suffix means that the value comes from the spi register.

The ramp timing is generated by the ramp state machine.

6.4.5.1. Ramp generators

On-chip ramp generator is provided supporting the multi-slot protocol of GSM/GPRS. Up to 4 consecutive bursts are supported as required by the multi-slot GPRS class 12.
The ramp profile itself is stored in a SRAM which can be programmed via the 4-wire serial interfaces, whereas the timing of the ramp profile is defined using the RU1[0], RU1[1], RU1[2], RU1[3] and RU1[4] 12-bit timing registers which can also be programmed via the SPI interface. These latter registers determine the valid data time, ramp-up time or ramp down time of each burst for the first APC DAC. Each SRAM consists of 160 10-bit words such that each ramp consists of 32 words.

Timing is related to the DAC enable signal, generated by the internal state handler. The rate of SRAM reading is programmable. Typically, in EGPRS mode, the SRAM reading is done in GSM quarter-bit rate: 13MHz/12 = 1.08333MHz, so each ramp cycle, which is 32-word long lasts 32 / 1.0833MHz = 29.5uS. Typically, in WCDMA mode, the SRAM is read at a 15.36MHz/12 rate. SRAM addresses are as follows:

- 0 to 31 : ramp-up #0 data
- 32 to 63 : ramp-up #1 data
- 64 to 95 : ramp-up #2 data
- 96 to 127 : ramp-up #3 data
- 128 to 159 : ramp-up #4 data.

Each address RU1[0] to RU1[4] controls the starting time of respectively ramp1 to ramp5. RU1[0] quarter-bit clocks after DAC enable rising edge the memory ramp1 will start to appear at the output of the APC DAC. After ramp1 has finished memory address 31 will be present, until RU1 quarter-bit clocks has passed from DAC enable rising edge. After that time this cycle will continue until ramp 5 is finished. If one RU register is set to 0 the respective ramp will be ignored and the memory pointer will go directly to the next ramp. If one RU register is smaller than the previous the correspondent ramp will be sent immediately after the previous ramp. Each time new data is written to the SRAM, the sequencer will be reset to the memory address zero.
The sequence starts after powering on the DAC enable signal, which is generated by the internal system control block. The timing controller counts the RU cycles until the start of a ramp action. During the ramp action 16 words are read from the SRAM at a half rate (2 QB time per word, which is 1.84615 uSec). The content of the 5 RU registers must fulfill the following requirement: RU1[i+1] > RU1[i] + 32. If using only n ramps (n<5) RU1[n+1] must be set to zero in order for the state machine to stop using the SRAM.
The following timing diagram illustrates the operation:

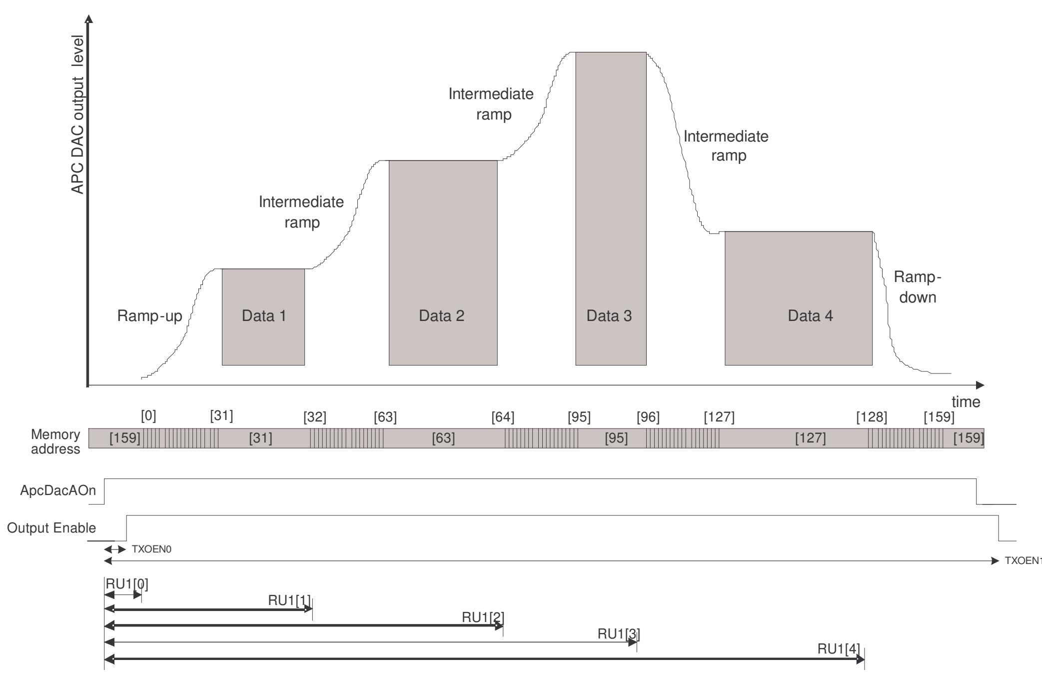

Figure 2 – Operation of the PA ramp generation

It is also possible to bypass the ramp generator and input the DAC values directly over the SPI.

The two DAC4 output signals are provided with pull-down resistors, which can be individually enabled. When in Ramp generation mode the enable of the DAC is controlled by the ramp state machine. Before TXOEN0 and TXOEN1 the DAC is disabled and the output will be pulled-down. These two delays can be adjusted via a SPI register.

The procedure to write/read into the RAM id the following:

- Set the initial address in register RAM_ADDR.
- Subsequent writes will start with initial address and increment after writing into the RAM.
- Read operation shares the same address register, so to read one must set the register RAM_ADDR with the initial address. After each reading the register will be incremented.
- If, for some reason, a write follows a read without setting RAM_ADDR, then the first read will be from the last written address plus 1.
- It is always possible to read the address that the state machine is putting on the RAM address input lines, be reading the register RAM_ADDR.

6.4.6. Power control signals

Each AuxDAC can be set independently into one of the following states:

- Power-on – all internal blocks are turned on
- Power-down – all internal blocks are turned off and power consumption is reduced to the minimum.

6.5. Clocks

6.5.1. PLLs

There are three PLLs available, all dedicated to the Rx path.

The clocking scheme is shown below:

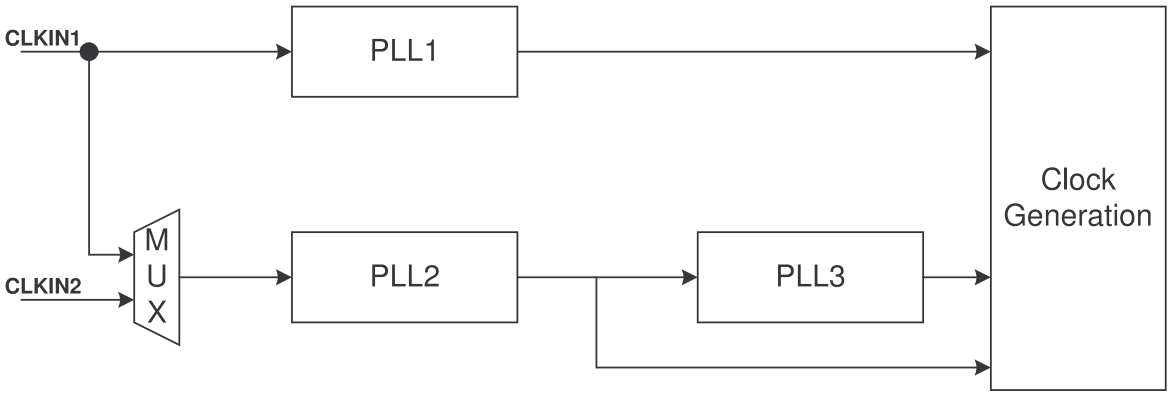

Figure 3 – Clock generation scheme

The Rx PLLs are used to generate the sampling clock for the sigma delta ADCs, digital filters and baseband interface. Each PLL contains three programmable dividers: pre-divider (**N**), feedback-divider (**M**) with a pre-divider by **1** or **2** and post-divider (**P**). Each divider can be programmed individually to load a divider ratio. Each PLL contains a lock detector, and its status can be read via the control registers.

The clock generation is performed in digital domain by a dedicated circuit.

In order to be compatible with all types of 2G-3G communication systems, the PLL system is optimized for three possible clock references: **13MHz**, **19.2MHz** and **26MHz**.

To generate **CDMA** clock frequency of **39.3216MHz** from **13MHz** and **26MHz** references, both PLL2 and PLL3 must be connected in series. For instance, PLL2 can generate **19.2MHz** from **13/26 MHz** reference, and then PLL3 generates **39.3216MHz** from **19.2MHz**.

The following table details the internal settings, the required input frequency and the expected output frequency for the possible selections of the dividers.

Input Clock (MHz)	Output Clock (MHz)	Prescale Factor (N-1)	Comparison freq. (MHz)	Multiplier (M-1)	Pre-Multiply by 2	VCO output (MHz)	Postscale Factor (P-1)	Overall Factor
19.2	***52.000***	**3**	6.40	**65**	**No**	416.00	**8**	2.7083
19.2	***39.3216***	**32**	0.60	**983**	**No**	589.80	**15**	2.0479
19.2	***61.440***	**3**	6.40	**48**	**No**	307.20	**5**	3.2000
26	***52.000***	**1**	26.00	**14**	**No**	364.00	**7**	2.0000
26	***61.440***	**65**	0.40	**768**	**No**	307.20	**5**	2.3631
26	***19.2***	**13**	2.00	**144**	**No**	288.00	**15**	0.7385
13	***52.000***	**1**	13.00	**28**	**No**	364.00	**7**	4.0000
13	***61.440***	**65**	0.20	**1536**	**No**	307.20	**5**	4.7262
13	***19.2***	**13**	1.00	**288**	**No**	288.00	**15**	1.4769

6.5.2. PLL Output Frequency

The VCO oscillating frequency is a function of the input reference frequency and of the multiplication/division ratios. It can be determined in the following way:

$$f_{VCO} = \frac{M}{N} \cdot f_{ref}$$

where M is the Feedback Multiplication Ratio and N is the Input Frequency Division Ratio.

However, some limits apply:

$$\frac{f_{ref}}{N} > 0.1MHz$$

$$250MHz < f_{VCO} < 600MHz$$

Finally, the Output Clock frequency is derived from a programmable division of the VCO frequency:

$$f_{out} = \frac{1}{P} \cdot f_{VCO}$$

For a certain M, N or P value it must be programmed **M-1, N-1** or **P-1** in the control pins. This is done through a SPI register.

6.5.3. Initialization

In order to facilitate the phase and frequency locking procedure, after power-up of the chip, the enable signal should be asserted for a short period (~**100ns**) and then de-asserted. This means that the PLL should be powered-on only at least **100ns** after the supply is stable.

The startup time is highly dependant on the chosen comparison frequency and can vary between **500us** for **400kHz** comparison and **16us** for **13MHz** comparison.

6.5.4. Power control and bypass signals

Each PLL can be set into one of the following states:

- <u>Enabled</u> – all internal blocks are turned on. Output clock is generated by the PLL.
- <u>Enabled & Bypass</u> – all internal blocks are turned on. Input clock is bypassed directly to the output of the PLL
- <u>Not enabled & Bypass</u> – all internal blocks are turned off and power consumption is reduced to the minimum. Input clock is bypassed directly to the output of the PLL.

6.5.4.1. PLL power-down mode

In power down mode, the control voltage of the VCO is actively pulled high, forcing the VCO oscillation to stop. Also, the Charge Pump current is cut off.
In power down mode, the output clock is the same as the PLL reference clock.
This mode also behaves as a reset sequence for the PLL, forcing it to start its lock acquisition from a low output frequency.

6.5.4.2. PLL bypass mode

This clock-multiplying PLL can be bypassed. When in bypass mode the input clock of the PLL is passed directly to the output without any processing. In this mode the PLL is still active.

6.6. Clock generation

Internally a phase generator receives the PLL clocks and generates all clocks needed for the data converters and digital logic operation, as well as the output clocks to the baseband processor.

Each PLL can be individually assigned to any of the RX blocks.

On each mode, if the division programmed is not sufficient to guarantee the data interface, the minimum output clock division is automatically selected.

6.7. References

All reference voltages and biasing currents needed for the Receive channel, Transmit channel, Clock generation block and Auxiliary converters are generated on-chip. Since these auxiliary blocks may be turned on during several signal conversions, special care is taken to guarantee that their power consumption is very low.

Each reference block can be in normal operation mode or in power-down mode. In this case total power consumption is less than **1µA**.

The operation modes are set through an SPI register.

6.7.1. Bandgap generator

A *bandgap*–based circuit generates a *bandgap* voltage needed as a stable reference (**±3.5%** variation) over the temperature range, to generate other reference voltages. This *bandgap* does not require any additional external decoupling capacitor.

6.7.2. LDO

An internal LDO is used to generate from the **2.7V** analog supply the **1.8V** supply need for IQ ADC, IQ DAC, AuxDACs and PLL operation. This LDO requires an additional external **1 uF** decoupling capacitor and is able to supply **75 mA** without significant drop–out.

The LDO power state can be controlled either through the SPI or through a dedicated pin (ENLDO). The following table summarizes the possible modes:

ENLDO (PIN)	SPI - REF_CFG<1>	LDO power state
0	0	Disabled
0	**1**	**Enabled**
1	0	**Enabled**
1	**1**	**Enabled**

By default (after a reset pulse is applied through RSTZ) the LDO is turned on through SPI, in order to ensure that all the blocks are able to start right after they are turned on.

To avoid current leakage in other blocks that are connected to the LDO, when in power-down the LDO output is forced to 0.

7. Data interface

7.1. Rx path

This device features two independent RX interfaces, but their timings are the same. Since the operating mode (EGPRS, CDMA and WCDMA) can differ, the receivers can operate simultaneously on distinct modes.

In order to minimize the number of interface pins between the AFE and the baseband modem, the I/Q output signals are multiplexed in the same data bus with the use of a control signal **RXIQ1/2**. This means that the actual **RXIQ1/2** frequency is two times the data interface sample rate.

Data transfer is synchronous with the PLL output clock **RXCLKOUT1/2.**

The following table describes the sample rate in both data interface and analog interface.

Operating mode	Sample rate (analog interface)	Sample rate (data interface)	Decimation ratio	***RXCLKOUT1/2*** **Clock frequency**
EGPRS	13 MHz	1083.(3) KHz (ZIF) 2166.(6) kHz (NZIF)	12	52 MHz 26 MHz 13 MHz 2.16(6) MHz
CDMA	39.3216 MHz	4.9152 MHz	8	39.3216 MHz 19.6608 MHz 9.8304 MHz
WCDMA	61.44 MHz	15.36 MHz	4	61.44 MHz 30.72 MHz

7.1.1. Typical data transfer

The next picture illustrates the timing interface for a typical EGPRS/CDMA/WCDMA transaction. The clock frequency can have more then 1 period for each **RXIQ1/2** signals, but in this case this signal indicates the correct sample rate and identifies the current sample as I or as Q. If clock rate is higher than required, the sample is hold until **RXIQ1/2** changes again

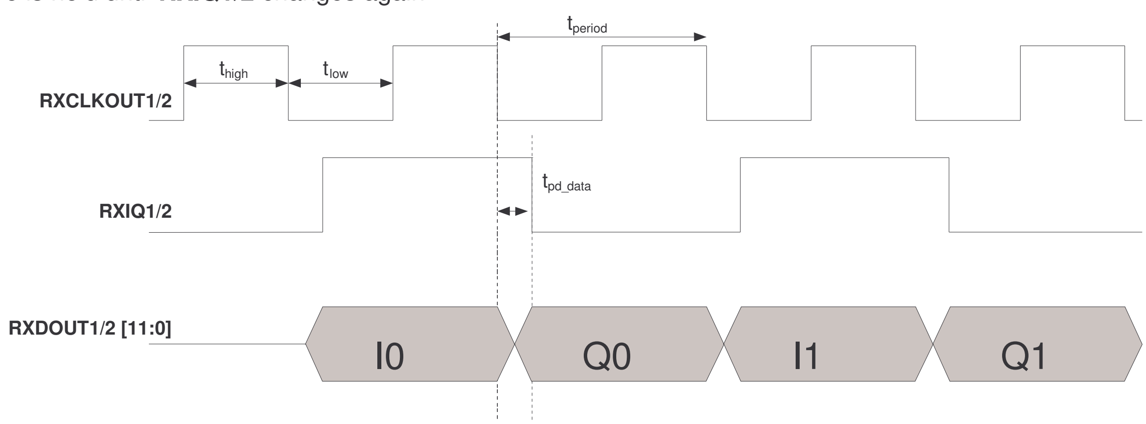

Figure 4 – Example of a typical RX transaction

7.1.2. Higher resolution transfers

In EGPRS, it is possible to transfer more than **12 bit** by splitting the word in two equal pieces and using a clock at double rate. MSBs are transmitted first. This mode of operation is described in the next picture.

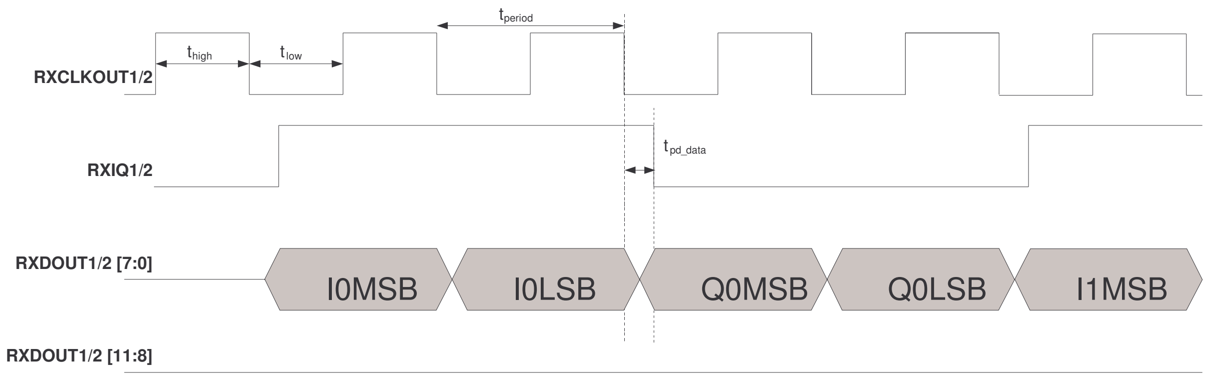

Figure 5 - Example of a double-rate RX transaction

On this interface mode only the data lines seven to zero are active, while the remaining ones are forced to zero. On the first clock cycle with HIGH on synchronization signal, the 8-MSB (most significant bits) of the I channel are sent, on the second period with synchronization signal HIGH the 8-LSB (least significant bits) of the I channel are sent. On the first clock cycle with synchronization signal LOW the Q MSB sample bits are transmitted and on the second cycle with synchronization signal LOW the Q LSB sample bits are transmitted.

The clock frequency can be higher than the double data rate. On this case only the first two samples for each **RXIQ1/2** should be sampled. This is shown in the next figure.

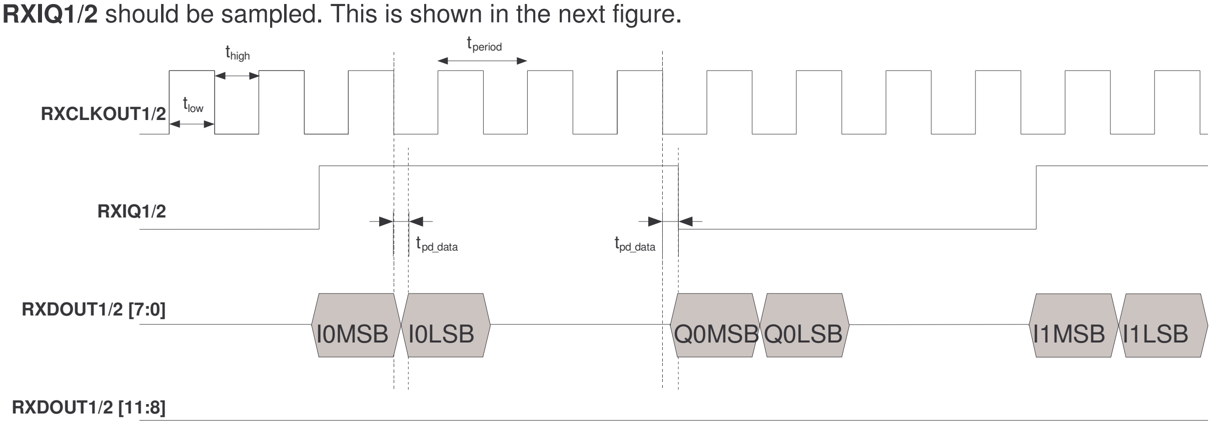

Figure 6 – Example if a double-rate RX transaction with higher clock

In ZIF mode, in order to output I and Q data at double-rate a minimum 4.33(3) MHz clock is required. When the interface clock is less than 4.33(3) MHz this mode cannot be used.

In NZIF mode, in order to output I and Q data at double-rate a minimum 8.66(6) MHz clock is required. When the interface clock is less than 8.66(6) MHz this mode cannot be used.

7.1.3. Timing requirements

The table below defines the timing requirements for single-edge clocking.

Symbol	Parameter	Min	Typ	Max	Unit
t_{period}	RXCLKOUT1/2 period (W-CDMA)	16			ns
t_{high}	RXCLKOUT1/2high period		50		%
t_{low}	RXCLKOUT1/2low period		50		%
t_{pd_data}	Output data propagation delay			5	ns

7.2. Tx path

In order to minimize the number of interface pins between the AFE and the baseband modem, the I/Q input signals are multiplexed in the same data bus with the use of a synchronization signal **TXIQ**. This means that the actual **TXIQ** frequency is two times the data interface sample rate.

Data transfer is fully synchronous with **TXCLK**, meaning that this interface is able to send data without the need of synchronization FIFOs.

The following table describes the sample rate in both data interface and analog interface.

Operating mode	Sample rate (data interface)	Sample rate (analog interface)	Interpolation ratio	*TXCLK* frequency
EGPRS	1083.(3) kHz [ZIF]	13 MHz	12	13 MHz
CDMA (mode I) **CDMA (mode II)** **CDMA (mode III)**	19.6608 MHz 9.8304 MHz 4.9152 MHz	39.3216 MHz	2 4 8	39.3216 MHz
WCDMA (mode I) **WCDMA (mode II)**	30.72 MHz 15.36 MHz	61.44 MHz	2 4	61.44 MHz

The next picture illustrates the timing interface for a typical EGPRS/CDMA/WCDMA transaction, where the data is sampled on the falling edge of the clock. The signal **TXIQ** when HIGH indicates an I sample and when LOW indicates a Q sample. The clock frequency can have more then 1 period for each **TXIQ** signal, but in this case this signal indicates the correct sample rate and identifies the current sample as I or as Q Since the clock always run at a higher rate, the data is sampled once after **TXIQ** changes value.

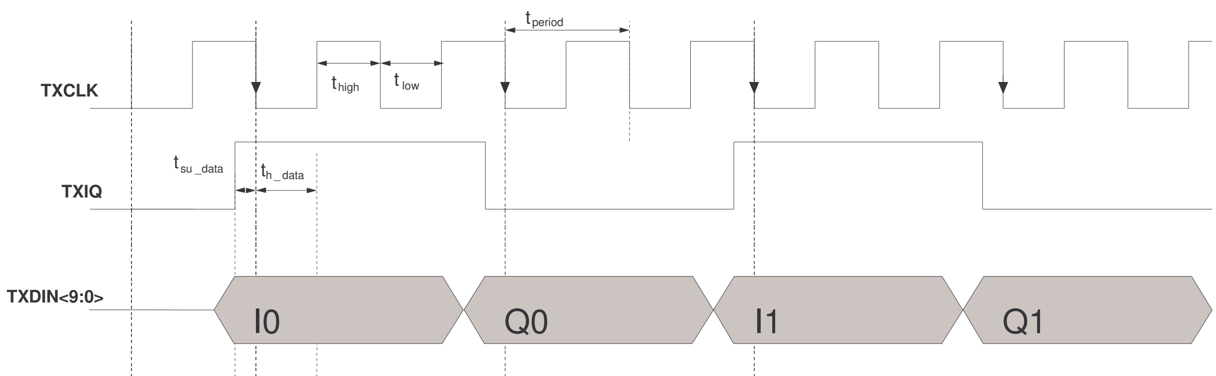

Figure 7 - Example of a typical TX transaction

7.2.1. Timing requirements

The table below defines the timing requirements.

Symbol	Parameter	Min	Typ	Max	Unit
t_{period}	TXCLK period (W-CDMA)	16			ns
t_{high}	TXCLK high period		50		%
t_{low}	TXCLK low period		50		%
t_{su_data}	Data setup time	5			ns
t_{h_data}	Data hold time	2			ns

7.2.2. IQDAC digital calibration

Calibration is activated by a SPI access. The references should be already stable. The process shall be finished after 2000 clock cycles. The calibration clock is derived directly from the TX clock as shown in the next table.

Interpolation ratio	*TXCLK* frequency
EGPRS	13 MHz
CDMA	19.6608 MHz
WCDMA	20.48 MHz

The timing diagram is shown below.

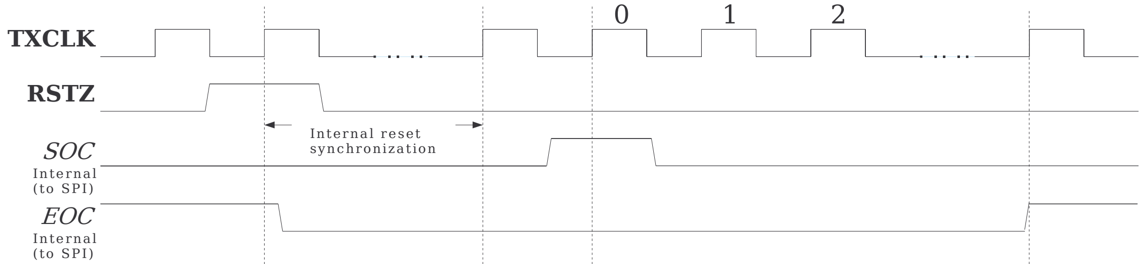

Figure 8 – IQDAC offset calibration cycle

7.3. AuxADC

This converter data transfer is controlled directly by a dedicated serial interface. Each conversion is started through a positive pulse on pin **AUXDACSOC**. The input mux selector (**SELAUXADCIN<7:0>** in the graphic) is defined by programming the appropriate register through the SPI.

The clock for the data conversion is generated externally and applied to the AFE through pin **CLKAUXADC**. Frequency must be lower than **4.8MHz,** regardless of the resolution mode.
Output data is updated in the rising edge of the clock.

The input signal of the ADC is sampled on the rising edge of clock. Latency of conversion is **12 cycles**.

The timing diagram is shown below.

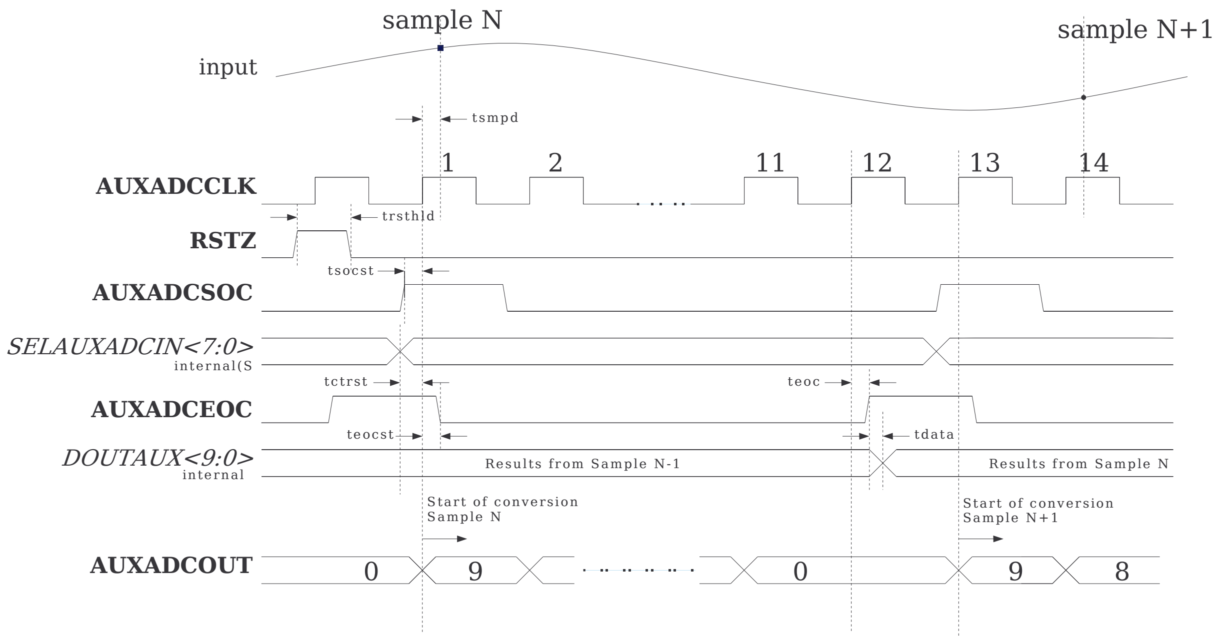

Figure 9 – AuxADC typical conversion cycle

7.3.1. Timing requirements

The table below defines the timing requirements.

Parameter	Conditions	MIN	TYP	MAX	Unit
Clock duty cycle		40	50	60	%
Data latency			12		cycles
clk falling edge to sampling delay (*tsmpd*)			24		ns
soc setup time before clk rise edge (*tsocst*)			3		ns
soc hold time after clk rise edge (*tsochld*)			3		ns
eoc rise time after clk rise edge (*teoc*)	@ 250 fF load			5.2	ns
Valid data out delay after eoc rise edge (*tdata*)	@ 250 fF load			7.0	ns

7.4. AuxDACs

The procedure is common for all AuxDACs, with the exception of the AuxDAC1.

7.4.1. AuxDAC2, AuxDAC3 and AuxDAC4

After the appropriate register is updated through the SPI with the data to be converted, a single clock cycle will be generated. Since the clock input is not needed for the internal operation of the DAC, only one cycle is enough to update the DAC. It is used only to capture the input data on register latches.

Input data is read in the falling edge of the clock. Output will be available after the falling edge.

The timing diagram is shown below:

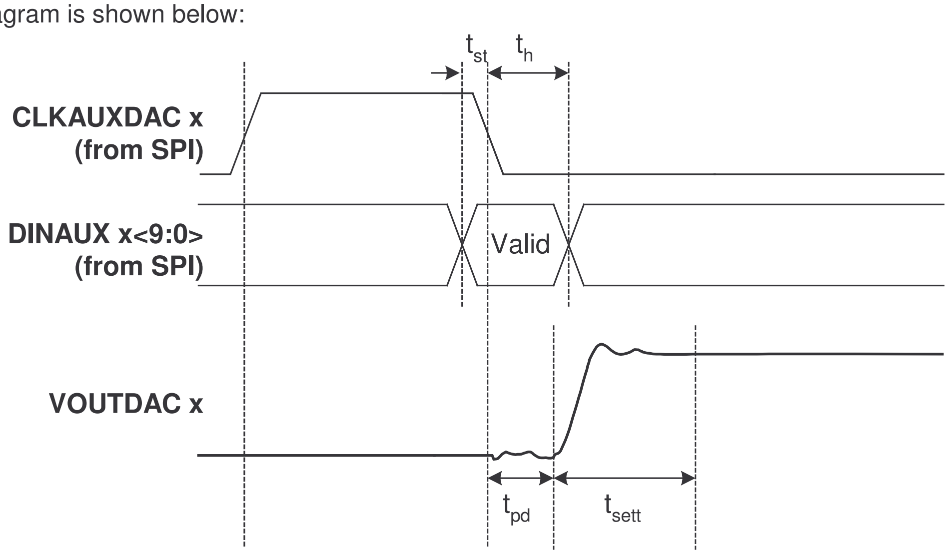

Figure 10 - Auxiliary DACs conversion cycle in normal operation mode

7.4.2. AuxDAC1

On this case, when the DAC is enabled the Sigma-delta clock is always enabled. The clock is divided from one of the Rx clocks and the division factor is programmable through a SPI register. The recommended division factor values are shown in the next table.

Operating mode	Division factor	RX clock	AuxDAC1 clock
EPGRS	8	52 MHz	6.50 MHz
CDMA	6	39.3216 MHz	6.55 MHz
WCDMA	9	61.44 MHz	6.83 MHz

To guarantee the required SINAD and DNL spec, the AuxDAC1 clock must be guaranteed to be always higher than **6.5MHz**.

The AuxDAC1 clock can be generated either from RX PLL1 or RX PLL2. This is selectable through the SPI.

7.4.3. Timing requirements

The table below defines the timing requirements.

Parameter	Conditions	MIN	TYP	MAX	Unit
Propagation delay (*tpd*)			5		ns
Settling time (*tsett+tpd*)	AFC DAC		40		µs
	AGC DAC		0.5		µs
	PA Control DAC		0.5		µs
	General purpose DAC		0.5		µs
Setup time for data in (*tst*)			0.5		ns
Hold time for data in (*th*)			5		ns

7.4.4. APCDAC ramp generator

The procedure is exactly same, with the only difference being that several write operations will be done during the ramp sequence, each one generating it's own clock sequence to update the APCDAC.

8. Serial-Parallel Interface (SPI)

All modes of operation are programmed through the SPI. This interface is able to be read, or be written even if the clock is absent outside the chip-select region.

The device samples and writes data always on the falling edge of the serial clock. When chip select is **LOW** a transaction is in progress. The falling edge of **SPICS** signals the start of the transaction and the rising edge indicates that the transaction is over. Address and data are transmitted with the most significant bit first. The SPI data maximum length can go up to 16-bit. The end of an SPI word is signaled by **SPICS**.

A typical timing diagram for a read operation is shown below for an example of an 8bit transaction.

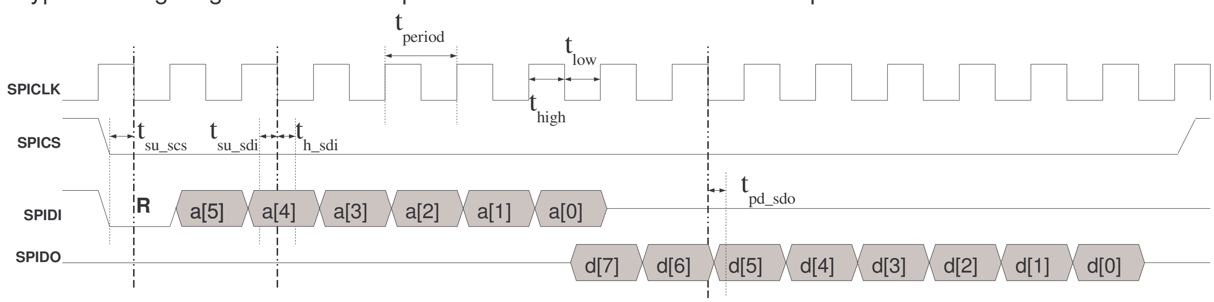

Figure 11 – SPI read operation

A typical timing diagram for a write operation is shown below for an example of an 8bit transaction.

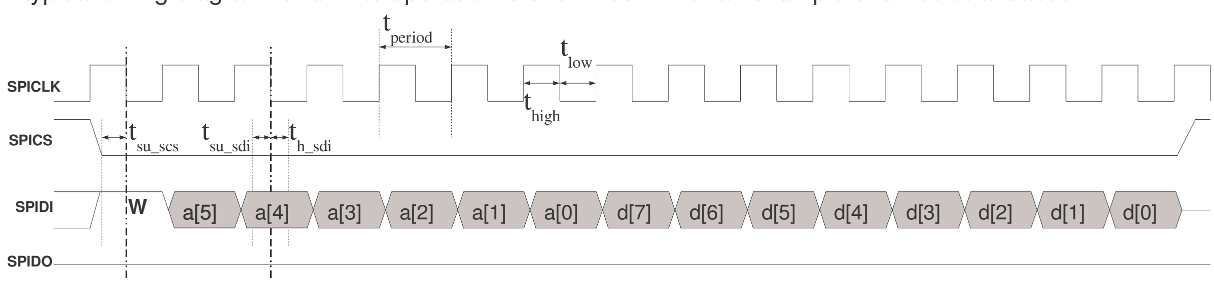

Figure 12 – SPI write operation

8.1.1. Timing requirements

The table below defines the timing requirements.

Symbol	Parameter	Min	Typ	Max	Unit
t_{period}	SPICLK period	50			ns
t_{high}	SPICLK high period		50		%
t_{low}	SPICLK low period		50		%
t_{su_SPICS}	SPICS setup time	5			ns
t_{su_SPIDI}	SPIDI setup time	5			ns
t_{h_SPIDI}	SPICS hold time	2			ns
t_{pd_SPIDO}	SPIDO propagation delay			10	ns

9. Interrupt mechanism

The AFE contains an **INT** pin which act as a level signal: if one of the internal interrupts is generated the line goes high. An internal status register contains flags for the various interrupt sources and can be read back by the baseband to find the source of the interrupt. When the baseband resets the interrupt source by programming the proper register, the **INT** line goes low again (provided all interrupt sources were dealt with). The various interrupt sources are the following:

1. Baseband - RXPLL 1 not locked
2. Baseband - RXPLL 2 not locked
3. Baseband - RXPLL 3 not locked

Each interrupt source can be separately masked programming the proper register

10. Registers

10.1. Address Map

The following list shows the address mapping of the various registers in the AFE.

Register Address	Register Name	Description	Read/Write
0x00	PLL1_CFG	PLL 1 configuration	Read/Write
0x01	PLL1_SCALE	PLL 1 pre and post scale values	Read/Write
0x02	PLL1_DIV	PLL 1 divider value	Read/Write
0x03	PLL2_CFG	PLL 2 configuration	Read/Write
0x04	PLL2_SCALE	PLL 2 pre and post scale values	Read/Write
0x05	PLL2_DIV	PLL 2 divider value	Read/Write
0x06	PLL3_CFG	PLL 3 configuration	Read/Write
0x07	PLL3_SCALE	PLL 3 pre and post scale values	Read/Write
0x08	PLL3_DIV	PLL 3 divider value	Read/Write
0x09	INT_STAT	Interrupt status	**Read only**
0x0A	INT_CLR	Interrupt latch clear	**Write only**
0x0B	INT_ENABLE	Interrupt source enable mask	Read/Write
0x0C	RX_CFG_1	RX paths configuration	Read/Write
0x0D	RX_CFG_2	RX paths configuration	Read/Write
0x0E	RX1_CFG	RX1 gain setup	Read/Write
0x0F	RX2_CFG	RX2 gain setup	Read/Write
0x10	TX_CFG	TX path configuration	Read/Write
0x11	TX_GAIN	TX path gain setup for voltage outputs	Read/Write
0x12	TX_CM	TX common-mode setup for voltage outputs	Read/Write
0x13	HK_CFG1	Housekeeping configuration 1	Read/Write
0x14	HK_CFG2	Housekeeping configuration 2	Read/Write
0x15	AFC_DAC_DAT	AFC DAC output data	Read/Write
0x16	AGC_DAC_DAT	AGC DAC output data	Read/Write
0x17	GP_DAC_DAT	GP DAC output data	Read/Write
0x18	APC_DAC_DAT	APC DAC output data	Read/Write
0x19	REF_CFG	AFE analog reference configuration	Read/Write
0x1A	RU[0]	PA ramp generator timing	Read/Write
0x1B	RU[1]	PA ramp generator timing	Read/Write
0x1C	RU[2]	PA ramp generator timing	Read/Write
0x1D	RU[3]	PA ramp generator timing	Read/Write
0x1E	RU[4]	PA ramp generator timing	Read/Write
0x1F	TXOEN0	Ramp generator delay	Read/Write
0x20	TXOEN1	Ramp generator delay	Read/Write
0x21	RAM_ADDR	Set ramp wrap write base address	Read/Write
0x22	RAM_WRITE	Write RAM and increment RAM address	Read/Write
0x23	PAD_CFG	Digital pads configuration	Read/Write
0x24	DEVICE	Device identification code	**Read only**
0x25	PAR_TEST	Parallel test mode	Read/Write
0x26	INTERNAL_TEST	Internal test mode	Read/Write
0x27	INTTEST_READ	Internal test mode read back	**Read only**
0x28	BIST_CTR	BIST controller	Read/Write
0x29	BIST_REG_MODE	BIST register	Read/Write

10.2. Register Details

The following table describes the content of the registers.
If some *reserved* mode is selected, the circuit will ignore it and stay at the same mode of operation.

Register Name	Power-up value	Bit	Name	Description
PLL1_CFG (0x00)	0x0	0	PLL1 Enable	0 = disable
				1 = enable
		1	PLL1 Bypass	0 = disable
				1 = enable
PLL1_SCALE (0x01)	0x0	7:0	PLL 1 Prescale	PLL1 pre-scale divider
		15:8	PLL 1 Postscale	PLL1 post-scale divider
PLL1_DIV (0x02)	0x0	0	PLL1 pre-divider	PLL1 feedback pre-divider by 2 (enable HIGH)
		11:1	PLL1 Divider	PLL1 feedback divider
PLL2_CFG (0x03)	0x0	0	PLL2 Enable	0 = disable
				1 = enable
		1	PLL2 Bypass	0 = disable
				1 = enable
		2	PLL2 Input Clock	0 = clkin1 / 1 = clkin2
PLL2_SCALE (0x04)	0x0	7:0	PLL2 Prescale	PLL2 pre-scale divider
		15:8	PLL2 Postscale	PLL2 post-scale divider
PLL2_DIV (0x05)	0x0	0	PLL2 pre-divider	PLL2 feedback pre-divider by 2 (enable HIGH)
		11:1	PLL2 Divider	PLL2 feedback divider
PLL3_CFG (0x06)	0x0	0	PLL3 Enable	0 = disable
				1 = enable
		1	PLL3 Bypass	0 = disable
				1 = enable
PLL3_SCALE (0x07)	0x0	7:0	PLL3 Prescale	PLL3 pre-scale divider
		15:8	PLL3 Postscale	PLL3 post-scale divider
PLL3_DIV (0x08)	0x0	0	PLL3 pre-divider	PLL3 feedback pre-divider by 2 (enable HIGH)
		11:1	PLL3 Divider	PLL3 feedback divider
INT_STAT (0x09)	0x0	0	PLL 1 Unlock	Set if PLL 1 is not locked
		1	PLL 2 Unlock	Set if PLL 2 is not locked
		2	PLL 3 Unlock	Set if PLL 2 is not locked
INT_CLR (0x0A)	0x0	0	PLL 1 Unlock	Set bit to clear interrupt
		1	PLL 2 Unlock	Set bit to clear interrupt
		2	PLL 3 Unlock	Set bit to clear interrupt
INT_ENABLE (0x0B)	0x0	0	PLL 1 Unlock	Set bit to enable interrupt source
		1	PLL 2 Unlock	Set bit to enable interrupt source
		2	PLL 3 Unlock	Set bit to enable interrupt source
RX_CFG_1 (0x0C)	0x0	0	RX1 power on	1 = on / 0 = off
		1	Rx2 power on	1 = on / 0 = off
		4:2	RX1 mode	000 = EGPRS ZIF
				001 = EGPRS NZIF
				010 = CDMA
				011 = WCDMA
				100 = *reserved*
				101 = EGPRS ZIF double rate
				110 = EGPRS NZIF double rate
				11x = *reserved*
		7:5	RX2 mode	000 = EGPRS ZIF
				001 = EGPRS NZIF
				010 = CDMA
				011 = WCDMA

				100 = *reserved* 101 = EGPRS ZIF double rate 110 = EGPRS NZIF double rate 11x = *reserved*
		9:8	RX1 output clock	00 = (52MHz EGPRS / 39.3216 Mhz CDMA / 61.44MHz WCDMA) 01 = (26MHz EGPRS / 19.6608 Mhz CDMA / 30.72MHz WCDMA) 02 = (13MHz EGPRS / 9.8304 Mhz CDMA / N.A. WCDMA) 03 = (2.16(6)MHz EGPRS / N.A. CDMA/ N.A WCDMA)
		11:10	RX2 output clock	00 = (52MHz EGPRS / 39.3216 Mhz CDMA / 61.44MHz WCDMA) 01 = (26MHz EGPRS / 19.6608 Mhz CDMA / 30.72MHz WCDMA) 02 = (13MHz EGPRS / 9.8304 Mhz CDMA / N.A. WCDMA) 03 = (2.16(6)MHz EGPRS / N.A. CDMA /N.A WCDMA)
		12	RX1 Input select	0 = external input 1 1 = external input 2
		14:13	RX2 Input select	00 = external input 3 01 = external input 4 10 = external input 5 11 = *reserved*
RX_CFG_2 (0x0D)	0x0	1:0	RX 1 input clock selection	00 = PLL 1 clock 01 = PLL2 clock 10 = PLL3 clock 11 = reserved
		3:2	RX 2 input clock selection	00 = PLL 1 clock 01 = PLL2 clock 10 = PLL3 clock 11 = reserved
RX1_CFG (0x0E)	0x0AA	1:0	ADC1 gain I input 1	00 = gain 0.5 in I for input 1 01 = gain 0.75 in I for input 1 10 = gain 1.0 in I for input 1 11 = gain 1.5 in I for input 1
		3:2	ADC1 gain I input 2	00 = gain 0.5 in I for input 2 01 = gain 0.75 in I for input 2 10 = gain 1.0 in I for input 2 11 = gain 1.5 in I for input 2
		5:4	ADC1 gain Q input 1	00 = gain 0.5 in Q for input 1 01 = gain 0.75 in Q for input 1 10 = gain 1.0 in Q for input 1 11 = gain 1.5 in Q for input 1
		7:6	ADC1 gain Q input 2	00 = gain 0.5 in Q for input 2 01 = gain 0.75 in Q for input 2 10 = gain 1.0 in Q for input 2 11 = gain 1.5 in Q for input 2
		8	RX1 clock delay enable	1 = on / 0 = off
		10:9	RX1 clock delay configuration	00 = 0ns 01 = T/8 (T=30.72MHz) 10 = 2*T/8 (T=30.72MHz) 11= 3*T/8 (T=30.72MHz)
RX2_CFG (0x0F)	0x0AAA	1:0	ADC2 gain I input 1	00 = gain 0.5 in I for input 3 01 = gain 0.75 in I for input 3 10 = gain 1.0 in I for input 3 11 = gain 1.5 in I for input 3
		3:2	ADC2 gain I input 2	00 = gain 0.5 in I for input 4

				01 = gain 0.75 in I for input 4 10 = gain 1.0 in I for input 4 11 = gain 1.5 in I for input 4
		5:4	ADC2 gain I input 3	00 = gain 0.5 in I for input 5 01 = gain 0.75 in I for input 5 10 = gain 1.0 in I for input 5 11 = gain 1.5 in I for input 5
		7:6	ADC2 gain Q input 1	00 = gain 0.5 in Q for input 3 01 = gain 0.75 in Q for input 3 10 = gain 1.0 in Q for input 3 11 = gain 1.5 in Q for input 3
		9:8	ADC2 gain Q input 2	00 = gain 0.5 in Q for input 4 01 = gain 0.75 in Q for input 4 10 = gain 1.0 in Q for input 4 11 = gain 1.5 in Q for input 4
		11:10	ADC2 gain Q input 3	00 = gain 0.5 in Q for input 5 01 = gain 0.75 in Q for input 5 10 = gain 1.0 in Q for input 5 11 = gain 1.5 in Q for input 5
		12	RX2 clock delay enable	1 = on / 0 = off
		14:13	RX2 clock delay configuration	00 = 0ns 01 = T/8 (T=30.72MHz) 10 = 2*T/8 (T=30.72MHz) 11= 3*T/8 (T=30.72MHz)
TX_CFG (0x10)	0x0	0	TX DAC power on	1 = on / 0 = off
		2:1	TX output mode	00 = no output, filter off 01 = voltage output, filter on 10 = current output, filter off 11 = *not used*
		5:3	TX mode	000 = EGPRS, interpolation 12x 001 = CDMA, interpolation 2x 010 = CDMA, interpolation 4x 011 = CDMA, interpolation 8x 100 = WCDMA, interpolation 2x 101 = WCDMA, interpolation 4x 11x =*reserved*
		6	TX current mode gain I	0 = gain 0 (Iout = 2mA for 250uA reference) 1 = gain 1 (Iout = 2.5mA for 250uA reference)
		7	TX current mode gain Q	0 = gain 0 (Iout = 2mA for 250uA reference) 1 = gain 1 (Iout = 2.5mA for 250uA reference)
		8	TX input current reference	0 = internal reference 1 = external reference (IREFTX)
		9	TX clock delay enable	1 = on / 0 = off
		11:10	TX clock delay configuration	00 = 0ns 01 = T/8 (T=30.72MHz) 10 = 2*T/8 (T=30.72MHz) 11= 3*T/8 (T=30.72MHz)
		12	TX start of calibration	0 = off / 1 = start calibration cycle. This is auto-clear. A read from this bit returns zero (provided DAC clock is runnning)
		13	TX end of calibration	0 = calibration is in process 1 = end of calibration cycle.
TX_GAIN (0x11)	0x0	3:0	TX I voltage output range	0000 = 0.88Vpp 0001 = 0.96Vpp

				0010 = 1.04Vpp 0011 = 1.12Vpp 0100 = 1.2Vpp 0101 = 1.28Vpp 0110 = 1.36Vpp 0111 = 1.44Vpp 1000 = 1.52Vpp 1001 = 1.6Vpp 1010 = 1.68Vpp 1011 = 1.76Vpp 1100 = 1.84Vpp 1101 = 1.92Vpp 1110 = 2Vpp 1111 = 2.08Vpp
		7:4	TX Q voltage output range	0000 = 0.88Vpp 0001 = 0.96Vpp 0010 = 1.04Vpp 0011 = 1.12Vpp 0100 = 1.2Vpp 0101 = 1.28Vpp 0110 = 1.36Vpp 0111 = 1.44Vpp 1000 = 1.52Vpp 1001 = 1.6Vpp 1010 = 1.68Vpp 1011 = 1.76Vpp 1100 = 1.84Vpp 1101 = 1.92Vpp 1110 = 2Vpp 1111 = 2.08Vpp
TX_CM (0x12)	0x0	2:0	Tx I voltage output common mode	000 = 1.0V common mode voltage 001 = 1.071V common mode voltage 010 = 1.143V common mode voltage 011 = 1.214V common mode voltage 100 = 1.286V common mode voltage 101 = 1.357V common mode voltage 110 = 1.429V common mode voltage 111 = 1.5V common mode voltage
		5:3	Tx Q voltage output common mode	000 = 1.0V common mode voltage 001 = 1.071V common mode voltage 010 = 1.143V common mode voltage 011 = 1.214V common mode voltage 100 = 1.286V common mode voltage 101 = 1.357V common mode voltage 110 = 1.429V common mode voltage 111 = 1.5V common mode voltage
HK_CFG1 (0x13)	0x0	0	AFC DAC enable	0 = disabled / 1 = enabled
		2:1	AFC DAC clock division factor	00 = 5 01 = 6 10 = 8 11 = 9
		4:3	AFC DAC clock source	00 = clk source is PLL1 01 = clk source is PLL2 10 = clk source is PLL3 11 = reserved

		5	AGC DAC enable	0 = disabled / 1 = enabled
		6	GP DAC enable	0 = disabled / 1 = enabled
		7	APC DAC enable	0 = disabled or enable controlled by the rampgen / 1 = enabled
		8	APC DAC ramp enable	0 = disabled / 1 = enabled
		9	Ramp burst mode	0 = ramp without delays RU1[i] 1 = use ramp timing delays RU1[i]
		11:10	APC DAC output select	00 = output disabled 01 = output A enabled 10 = output B enabled 11 = *not used*
		13:12	APC DAC output pull down	00 = pull-down disable 01 = output A pull-down enabled 10 = output B pull-down enabled 11 = both outputs pull-down enabled
HK_CFG2 (0x14)	0x0	0	ADC enable	0 = disabled / 1 = enabled
		4:1	ADC resolution select	0000 = 10bit, 400kS/sec 0001 = 9bit, 436.36kS/sec 0010 = 8bit, 480kS/sec 0011 = 7bit, 533.33kS/sec 0100 = 6bit, 600kS/sec 0101 = 5bit, 685.71kS/sec 0110 = 4bit, 800kS/sec 0111 = 3bit, 960kS/sec 1000 = 2bit, 1200kS/sec 1xx1= *reserved* 1x1x= *reserved* 11xx= *reserved*
		7:5	ADC input select	000 = external input 1 001 = external input 2 010 = external input 3 011 = external input 4 100 = external input 5 101 = external input 6 110 = *reserved* 111 = *reserved*
		8	ADC reference mode	0 = internal reference (AVDD 2.7V) 1 = external reference (VREFAUX)
AFC_DAC_DAT (0x15)	0x0	11:0	AFC DAC input data	12-bit data
AGC_DAC_DAT (0x16)	0x0	9:0	AGC DAC input data	10-bit data
GP_DAC_DAT (0x17)	0x0	9:0	GP DAC input data	10-bit data
APC_DAC_DAT (0x18)	0x0	9:0	APC DAC input data	10-bit data
REF_CFG (0x19)	0x1	0	Bandgap generator enable	0 = disabled / 1 = enabled
		1	LDO enable	0 = disabled / 1 = enabled
RU[0] (0x1A)	0x0	11:0	Delay for first ramp on APC DAC Time unit: EGPRS=1/(1.083MHz) WCDMA=1/(1.28MHz)	0 = 0 sec / 0xfff = delay is the same as ramp duration
RU[1] (0x1B)	0x0	11:0	Delay for second ramp on APC DAC Time unit: EGPRS=1/(1.083MHz) WCDMA=1/(1.28MHz)	0 = 0 sec 0xfff = delay is the same as ramp duration
RU[2] (0x1C)	0x0	11:0	Delay for third ramp on APC DAC Time unit: EGPRS=1/(1.083MHz) WCDMA=1/(1.28MHz)	0 = 0 sec 0xfff = delay is the same as ramp duration
RU[3] (0x1D)	0x0	11:0	Delay for fourth ramp on APC DAC Time unit: EGPRS=1/(1.083MHz) WCDMA=1/(1.28MHz)	0 = 0 sec 0xfff = delay is the same as ramp duration
RU[4] (0x1E)	0x0	11:0	Delay for fifth ramp on APC DAC Time unit: EGPRS=1/(1.083MHz) WCDMA=1/(1.28MHz)	0 = 0 sec 0xfff = delay is the same as ramp duration

TXOEN0 (0x1F)	0x0	11:0	Delay to output on Time unit: EGPRS=1/(1.083MHz) WCDMA=1/(1.28MHz)	0 = 0 sec 0xfff = delay is the same as ramp duration
TXOEN1 (0x20)	0x0	11:0	Delay to output off Time unit: EGPRS=1/(1.083MHz) WCDMA=1/(1.28MHz)	0 = 0 sec 0xfff = delay is the same as ramp duration
RAM_ADDR (0x21)	0x0	7:0	RAM address	Sets the address for the RAM auto-increment write process. A write to the RAM will auto-increment the write address pointer
RAM_WRITE (0x22)	0x0	9:0	RAM data	Data to be written into the RAM, starting at the base address with the address auto-incrementing
PAD_CFG (0x23)	0x0	0	Drive strength interrupt output	0 = 4mA; 1 = 8mA
		1	Drive strength RX1 outputs	0 = 4mA; 1 = 8mA
		2	Drive strength RX1 clock	0 = 4mA; 1 = 8mA
		3	Drive strength RX2 outputs	0 = 4mA; 1 = 8mA
		4	Drive strength RX2 clock	0 = 4mA; 1 = 8mA
		5	Drive strength AUXADC outputs	0 = 4mA; 1 = 8mA
		6	Drive strength SPI output	0 = 4mA; 1 = 8mA
DEVICE (0x24)	0x0	3:0	Version	Version number
		11:4	Part	Part number
PAR_TEST (0x25)	0x0	2:0	Parallel TX test modes	000 – TX in normal mode 001 – TX Direct access to one analog channel 010 – TX direct access to both analog channels 011 – TX bypass equalizer 100 – TX bypass interpolator
		3	Parallel RX1 test modes	0 – RX1 in normal mode 1 – RX 1 direct access to analog channel
		4	Parallel RX2 test modes	0 – RX2 in normal mode 1 – RX2 direct access to analog channel
		6:5	AUX DAC 1 test modes	00 – DAC 1 in normal mode 01 – DAC 1 direct access to analog core 10 – DAC 1 sigma-delta plus analog data converter test mode 11 = reserved
		7	AUX DAC 2 test modes	0 – DAC 2 in normal mode 1 – DAC 2 in test mode
		8	AUX DAC 3 test modes	0 – DAC 3 in normal mode 1 – DAC 3 in test mode
		9	AUX DAC 4 test modes	0 – DAC 4 in normal mode 1 – DAC 4 in test mode
		10	AUX ADC test modes	0 – AUX ADC in normal mode 1 – AUX ADC in test mode
		11	PLL 1 test mode	0 – PLL1 in normal mode 1 – PLL1 in test mode
		12	PLL 2 test mode	0 – PLL2 in normal mode 1 – PLL2 in test mode
		13	PLL 3 test mode	0 – PLL3 in normal mode 1 – PLL3 in test mode
INTERNAL_TEST (0x26)	0x0	7:0	Test mode data	Test mode data to program internal test registers
		13:8	Test mode address	Test mode address to program internal rest registers
INTTEST_READ (0x27)	0x0	15:0	Test data output	Test data output according to the setting of INTERNAL_TEST register
BIST_CTR (0x28)	0x3F0	0	BIST mode	When high sets the device to BIST mode
		1	Load mode register	Write to program the BIST_REG_MODE
		2	Start memory BIST	Write one to start the BIST (the bist mode must be seted)
		3	BIST on	Read-only, when high the BIST is operating
		9:4	BIST upper mode register	BIST memory diagnostics
		13:10	Selects the output debug memory	Output selected RAM debug information: 0000 – mem 0 LSB / 1000 – mem 0 MSB

				0001 – mem 1 LSB / 1001 – mem 1 MSB 0010 – mem 2 LSB / 1010 – mem 2 MSB 0011 – mem 3 LSB / 1011 – mem 3 MSB 0100 – mem 4 LSB / 1100 – mem 4 MSB 0101 – mem 5 LSB / 1101 – mem 5 MSB x11x – reserved
BIST_REG_MODE (0x29)	0x3C	14:0	BIST mode register	Sets the BIST operating mode

11. Parallel Test Modes

All analog blocks have external parallel access to the internal digital interfaces to allow high-speed functional and parametric testing. Data shall be accessed through the parallel interface using a synchronous clock. Each mode is configured through a SPI register (PAR_TEST).

Warning: only one device at a time can be programmed to test mode. If more than one device is programmed to enter in test-mode, it will be considered invalid and all the devices will be forced to normal mode..

Test mode details are shown in the following sub-sections.

11.1. TX channel parallel test modes

11.1.1. Direct access to I DAC or Q DAC inputs

In this mode, DAC input data is applied directly to each TX channel.

Test mode configuration

- **DAC1:** PAR_TEST=0x0001

I/O pins

- DAC Clock: **TXCLK**
- DAC I - Data<9:0>: **TXDIN<9:0>** if **TXIQ='1'**
- DAC Q - Data<9:0>: **TXDIN<9:0>** if **TXIQ='0'**

The signal timings are shown in the next figure:

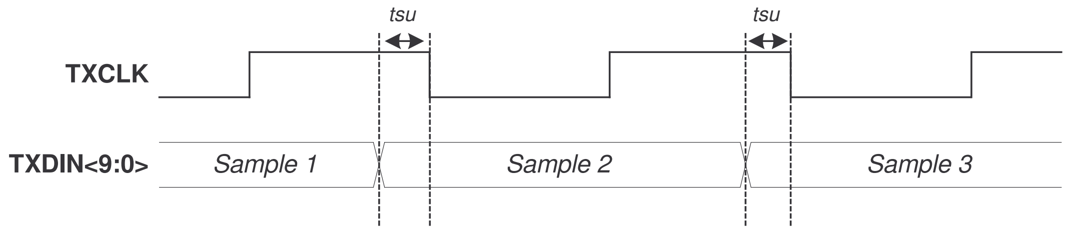

Figure 13 – Timing diagram for TX test mode

11.1.2. Direct access to both I & Q DACs inputs

In this mode, DAC input data is applied directly to both TX I and Q at the same time.

Test mode configuration

- **DAC1 & DAC2:** PAR_TEST=0x0002

I/O pins

- DAC Clock: **TXCLK**
- DAC I - Data<9:0>: **TXDIN<9:0>**
- DAC Q - Data<9:0>: **TXDIN<9:0>**
- DAC enable: **TXIQ**

The signal timings are shown in the next figure:

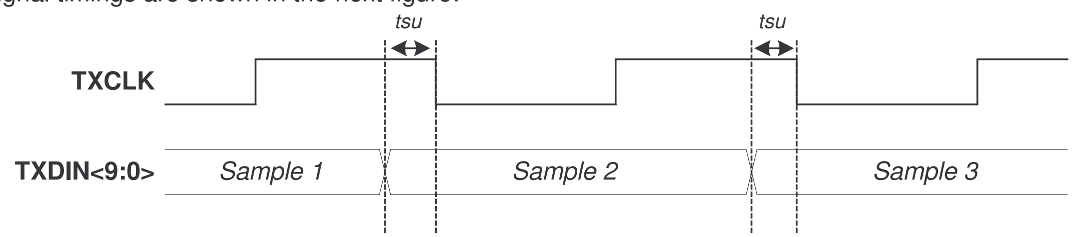

Figure 14 – Timing diagram for TX test mode

Minimum setup time t_{su} is **1ns**.

11.1.3. Bypass digital gain equalizer

In this mode, digital gain equalizer is bypassed and the input data is applied directly to the Interpolator.

Test mode configuration

- **DAC1:** PAR_TEST=0x0003

This mode works exactly like in normal mode of operation. The signal timings defined on section 7.2 must be followed.

11.1.4. Bypass interpolator

In this mode, digital interpolator is bypassed and the input data, after passing the digital gain equalizer. is applied directly to the Interpolator.

Test mode configuration

- **DAC1:** PAR_TEST=0x0004

This mode works exactly like in normal mode of operation. The signal timings defined on section 7.2 must be followed.

11.2. RX channel parallel test modes

11.2.1. Direct access from RX1 ADC & RX2 ADC outputs

In this mode, input data is available directly from the RX outputs.

Test mode configuration

- **RX1:** PAR_TEST=0x0008
- **RX2:** PAR_TEST=0x0010

I/O pins

- RX1&2 Clock: **TXCLK**
- RX1 I - Data<3:0>: **RXDOUT1<3:0>**
- RX1 Q - Data<3:0>: **RXDOUT1<7:4>**
- RX2 I - Data<3:0>: **RXDOUT2<3:0>**
- RX2 Q - Data<3:0>: **RXDOUT2<7:4>**

The signal timings are shown in the next figure:

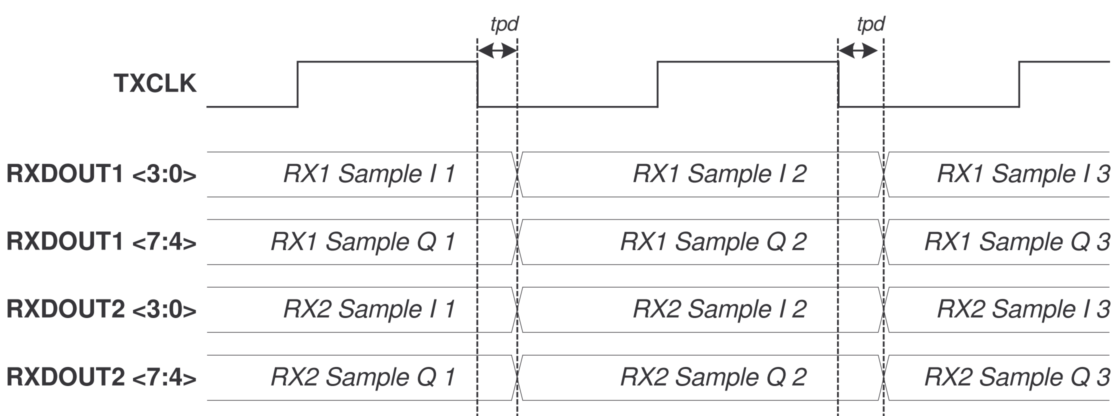

Figure 15 - Timing diagram for RX test modes

Propagation delay t_{pd} is less than **2ns**.

11.3. Auxiliary converters parallel test modes

11.3.1. Direct access to AUXDAC inputs

In this mode, input data is read directly from the AuxDAC outputs.

Test mode configuration

- **DAC1:** PAR_TEST=0x0020
- **DAC2:** PAR_TEST=0x0080
- **DAC3:** PAR_TEST=0x0100
- **DAC4:** PAR_TEST=0x0200

Only one DAC can be in test mode at a time. The enabled one receives directly from the chip pads the clock and the data to be converted:

I/O pins

- Clock: **TXCLK**
- Data<11:0>: **TXDIN<9:0>**
- Enable: **TXIQ**

The signal timings defined on section 7.4 must be followed. Additionally, for AuxDAC4 these signals are also available (output pull down is disabled)

- Output enable bit 1: **AUXADCSOC**
- Output enable bit 0: **AUXADCCLK**

11.3.2. Direct access to AUXDAC1 sigma-delta input inputs

In this mode, input data is applied directly to the AuxADC1 sigma-delta input.

Test mode configuration

- **AuxDAC1+Sigma-Delta:** PAR_TEST=0x0040

I/O pins

- Clock: **TXCLK** (data is latched on the rising edge of this clock signal)
- Data<11:0>: **TXDIN<9:0>**, **AUXADCCLK**, **AUXADCSOC**
- Enable: **TXIQ**

The signal timings are shown in the next figure:

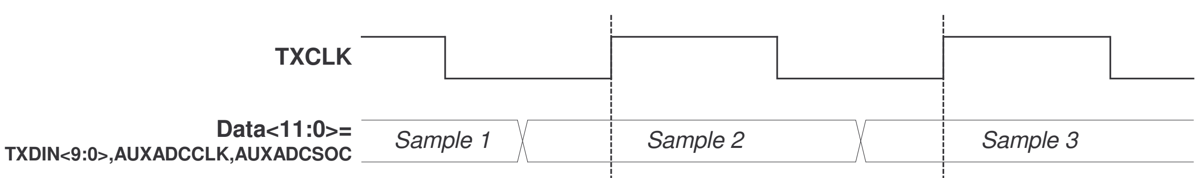

Figure 16 – Timing diagram for AuxDAC1 test mode

11.3.3. Direct access from Auxiliary ADC outputs inputs

On this mode the AuxADC outputs are available on the following pads:

Test mode configuration

- **AuxADC1:** PAR_TEST=0x0400

I/O pins

- Clock: **AUXADCCLK** (same as normal mode)
- Start of conversion: **AUXADCSOC** (same as normal mode)
- End of conversion: **AUXADCEOC** (same as normal mode)
- Data<9:0>: **RXDOUT1<9:0>**
- Input selector: **TXDIN<2:0>**
- Reference selector: **TXDIN<3>**

- Resolution selector: **TXDIN<7:4>**
- Enable**: TXIQ**

The signal timings defined on section 7.3 must be followed.

11.4. Direct access to PLL outputs

In this test mode, PLL1 clock signal is routed to **RX1CLKOUT1** and PLL2 or PLL3 clock signals are routed to **RX1CLKOUT2**. In order to minimize the jitter injected by the digital circuitry, the signal path in the digital core has high strength buffers and the routing is isolated from other internal signals. The logic to the output consists of a simple multiplexer.

Test mode configuration

- **PLL1:** PAR_TEST=0x0800
- **PLL2:** PAR_TEST=0x1000
- **PLL3:** PAR_TEST=0x2000

12. Production Test

12.1. Scan test

All digital logic has a standard scan test to meet better than 97% coverage. There are a total of 8 chains for the Baseband section.

To enter scan mode, the pin SCANMODE must be forced HIGH. When in this mode, some of the pins will have different assignments:

12.2. BIST for RAMs

The RAM blocks are tested through a BIST interface.

The BIST interface is accessible using the register file BIST registers. There are two registers:

- **BIST_CRT**: Sets BIST mode, load command to load the BIST register mode, triggers the BIST, controls the upper part of the mode register model and for debug purposes selects the result of one the memories;
- **BIST_REG_MODE:** Configures the operation of the BIST.

In BIST, the clock frequency is 10MHz.

The sequence to program a BIST mode is as follows:

- Set the device to BIST mode. From this point on the SPI clock will be used as BIST clock (the SPI clock should be always available)
- Set up correctly BIST register mode (this requires writing to BIST_REG_MODE, followed by an all ones write into BIST_CTR – upper part of the register mode, and then writing into the bit load mode register in BIST_CTR)
- After configuring the BIST, write into BIST_CTR – start memory BIST

When in BIST mode some of the pins are multiplexed (the SPI signals are operating normally):

By multiplexing these pins, the BIST result can be monitored using the parallel interface, which will increase the efficiency of the tester.

13. References decoupling

The following decoupling is recommended for the correct operation of the reference circuits and to guarantee the specified performance in the AFE conversion blocks.

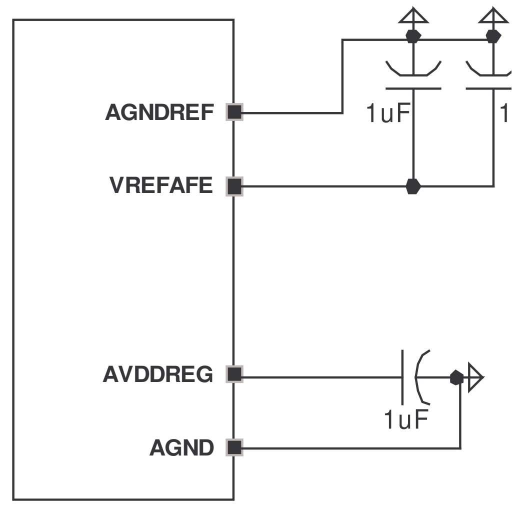

Figure 17 – Suggested decoupling of reference pins